

Exploratory Analysis

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Load in Packages

```
# load in necessary packages  
library("tidyverse")
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.1.4      v readr      2.1.5  
v forcats    1.0.0      v stringr    1.5.1  
v ggplot2     4.0.2      v tibble     3.2.1  
v lubridate   1.9.4      v tidyr      1.3.1  
v purrr       1.0.4  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library("brms")
```

```
Loading required package: Rcpp  
Loading 'brms' package (version 2.23.0). Useful instructions  
can be found by typing help('brms'). A more detailed introduction  
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':
```

```
ar
```

```
library("ggeffects")
library("readxl")
library("RColorBrewer")
```

Read in Data

```
# read in the data from the excel file
Poll_Data <- read_xlsx("Ellie_ChiSquare.xlsx", sheet = 1)
```

Warning: Expecting numeric in A1192 / R1192C1: got 'male'

```
# make sure the data file is in correctly
head(Poll_Data)
```

```
# A tibble: 6 x 28
  index UniqueID DateCollected Site Collector Method PollinatorOrder
  <dbl>   <dbl> <dtm>          <chr>   <chr>    <chr> <chr>
1 876     956 2021-07-11 00:00:00 Chadwick Jenny Hsu   Netti~ Hymenoptera
2 893    1021 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
3 892    1020 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
4 902    1030 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
5 900    1028 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
6 896    1024 2021-07-11 00:00:00 Chadwick Sofie Andr~ Netti~ Hymenoptera
# i 21 more variables: PollinatorGroupGeneral <chr>, PollinatorFamily <chr>,
# PollinatorTribe <chr>, PollinatorGenus <chr>, PollinatorSpecies <chr>,
# PollGenusSpeciesUSE <chr>, PollinatorSex <chr>, FlowerCollectedOn <chr>,
# PlantType <chr>, PlantFamily <chr>, PlantGenus <chr>, PlantSpecies <chr>,
# PlantGenusSpeciesUSE <chr>, Native <chr>, FlowerColor <chr>,
# SingleFlColor <chr>, FlowerShape <chr>, AllPollBeeOnly <chr>,
# PollenNectar <chr>, PinnedBy <chr>, Notes <chr>
```

Flower Shape Dataframes

```
# Create new dataframe
Fl_Shape <- Poll_Data %>%
  # used the select function to keep only 4 columns
  select(PollinatorOrder, PollinatorFamily, PollGenusSpeciesUSE, FlowerShape) %>%
```

```
# Get rid of flowers with unknown shape
filter(FlowerShape != "U")

# create new dataframe for just bees
Fl_ShapeSp <- Fl_Shape %>%
  # use mutate to create new column to separate honey bees from wild bees
  mutate(BeeType = ifelse(Fl_Shape$PollGenusSpeciesUSE == "Apis mellifera",
    "Honey Bee",
    "Wild Bee")) %>%
  # Filter dataframe to just include hymenoptera and get rid of non-bees
  filter(PollinatorOrder == "Hymenoptera", PollinatorFamily != "Agridae")

# check it filtered the families correctly
print(unique(Fl_ShapeSp$PollinatorFamily))
```

```
[1] "Apidae"      "Halictidae"  "Megachilidae" "Colletidae"
```

```
# remove unnecessary columns
Fl_Shape_Order <- Fl_Shape %>%
  select(PollinatorOrder, FlowerShape)

# data frame of 2 flower shapes I want to focus on
Fl_Shape_Final <- Fl_Shape_Order %>%
  group_by(PollinatorOrder, FlowerShape) %>%
  summarize(n = n()) %>%
  mutate(total = sum(n),
    prop = n / sum(n)) %>%
  ungroup() %>%
  filter(FlowerShape == "Dish" | FlowerShape == "Tube")
```

`summarise()` has grouped output by 'PollinatorOrder'. You can override using the `.groups` argument.

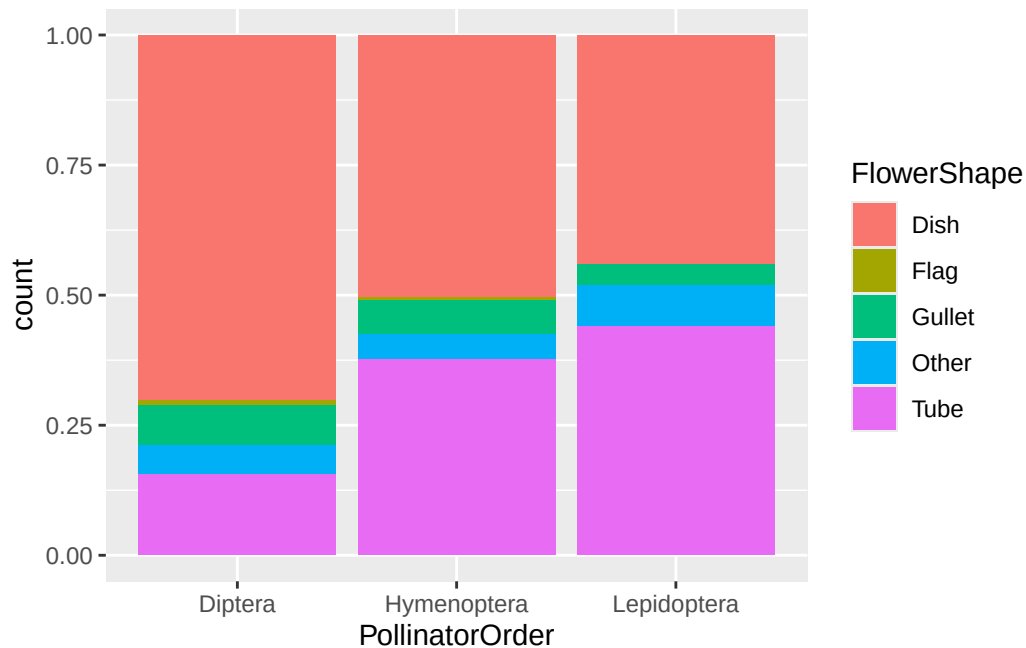
```
Fl_Shape_Final
```

```
# A tibble: 6 x 5
  PollinatorOrder FlowerShape      n total  prop
  <chr>           <chr>    <int> <int> <dbl>
1 Diptera        Dish      163   232 0.703
```

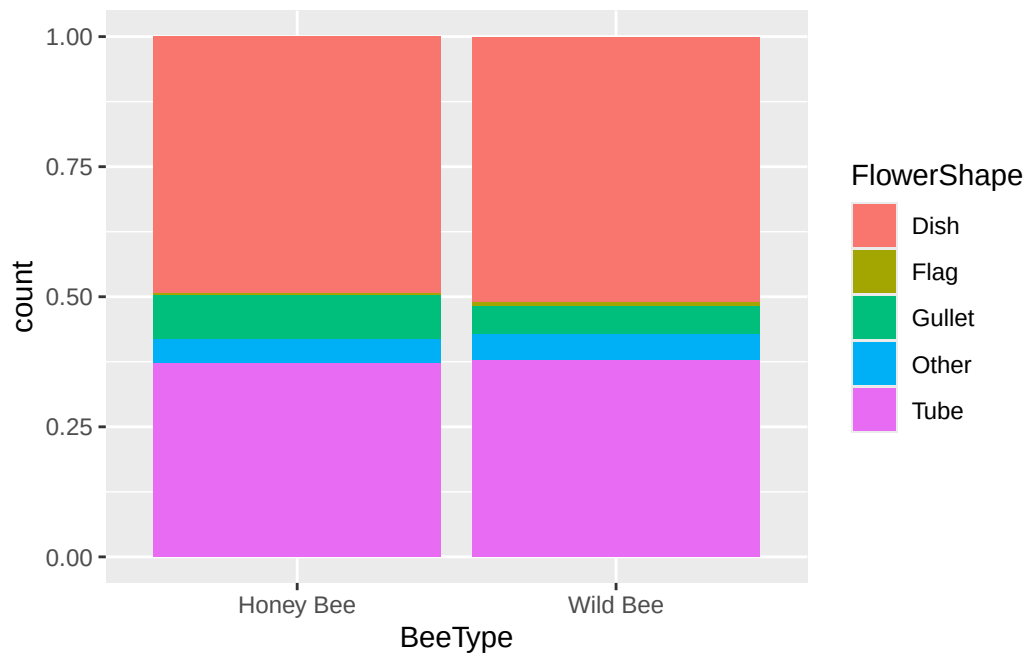
2	Diptera	Tube	36	232	0.155
3	Hymenoptera	Dish	563	1118	0.504
4	Hymenoptera	Tube	421	1118	0.377
5	Lepidoptera	Dish	22	50	0.44
6	Lepidoptera	Tube	22	50	0.44

Exploratory Graphs

```
# Exploratory Graph based on Order and FLower Shape
ggplot(Fl_Shape, aes(PollinatorOrder, fill = FlowerShape)) +
  # make a stacked bar plot to see differences in percent visitation frequency among orders
  geom_bar(position = "fill")
```



```
# Exploratory Graph based on Bee Type and FLower Shape
ggplot(Fl_ShapeSp, aes(BeeType, fill = FlowerShape)) +
  # make a stacked bar plot to see differences in percent visitation frequency among orders
  geom_bar(position = "fill")
```



Contingency Tables

```
# Create contingency table for shape and pollinator order for easier analysis
ShapeOrder <- table(
  Fl_Shape$FlowerShape, Fl_Shape$PollinatorOrder)
# print to make sure it works
print(ShapeOrder)
```

	Diptera	Hymenoptera	Lepidoptera
Dish	163	563	22
Flag	2	7	0
Gullet	18	73	2
Other	13	54	4
Tube	36	421	22

```
# easy frequentist statistical analysis to see potential alongside graph
a <- chisq.test(ShapeOrder)
```

Warning in chisq.test(ShapeOrder): Chi-squared approximation may be incorrect

```
print(a)
```

Pearson's Chi-squared test

```
data: ShapeOrder
X-squared = 46.665, df = 8, p-value = 1.776e-07
```

```
# Create contingency table for shape and Bee Type for easier analysis
ShapeBee <- table(
  Fl_ShapeSp$FlowerShape, Fl_ShapeSp$BeeType)
# print to make sure it works
print(ShapeBee)
```

	Honey Bee	Wild Bee
Dish	217	346
Flag	2	5
Gullet	37	36
Other	20	34
Tube	164	256

```
# easy frequentist statistical analysis to see potential alongside graph
b <- chisq.test(ShapeBee)
```

Warning in chisq.test(ShapeBee): Chi-squared approximation may be incorrect

```
print(b)
```

Pearson's Chi-squared test

```
data: ShapeBee
X-squared = 4.5588, df = 4, p-value = 0.3356
```

Flower Color Dataframes

```
# Create new dataframe
Fl_Color <- Poll_Data %>%
  # used the select function to keep only 4 columns
  select(PollinatorOrder, PollinatorFamily, PollGenusSpeciesUSE, SingleFlColor) %>%
  # Get rid of flowers with unknown color
  filter(SingleFlColor != "U", SingleFlColor != "None")

print(unique(Fl_Color$SingleFlColor))
```

```
[1] "Blue"    "Purple" "Green"   "Orange" "Red"     "Yellow" "Pink"    "White"
```

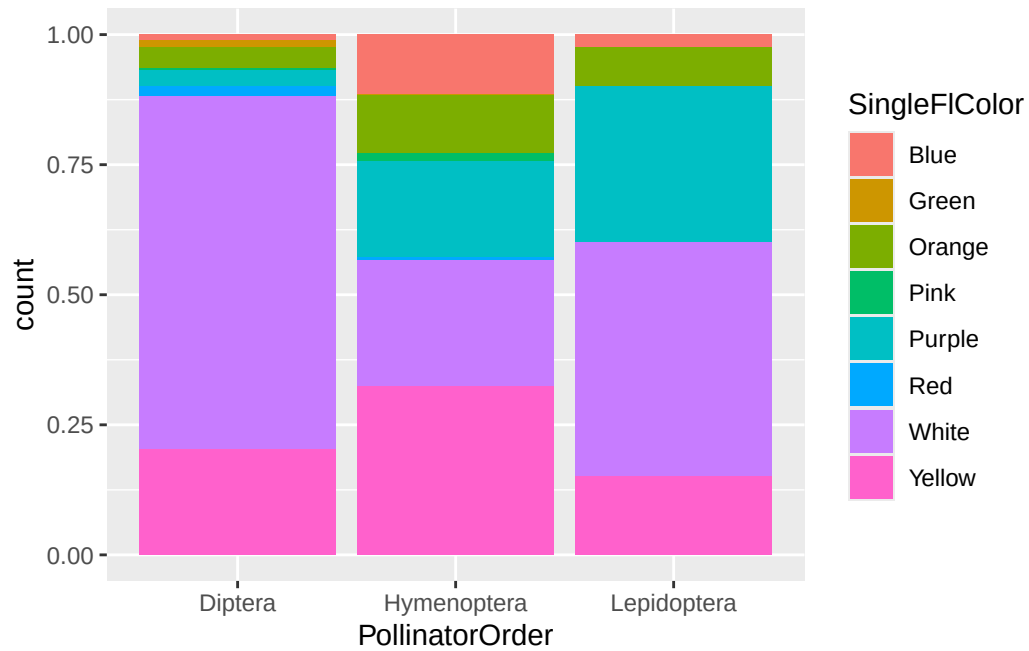
```
# create new dataframe for just bees
Fl_ColorSp <- Fl_Color %>%
  # use mutate to create new column to separate honey bees from wild bees
  mutate(BeeType = ifelse(Fl_Color$PollGenusSpeciesUSE == "Apis mellifera",
    "Honey Bee",
    "Wild Bee")) %>%
  # Filter dataframe to just include hymenoptera and get rid of non-bees
  filter(PollinatorOrder == "Hymenoptera", PollinatorFamily != "Agridae")

# check it filtered the families correctly
print(unique(Fl_ColorSp$PollinatorFamily))
```

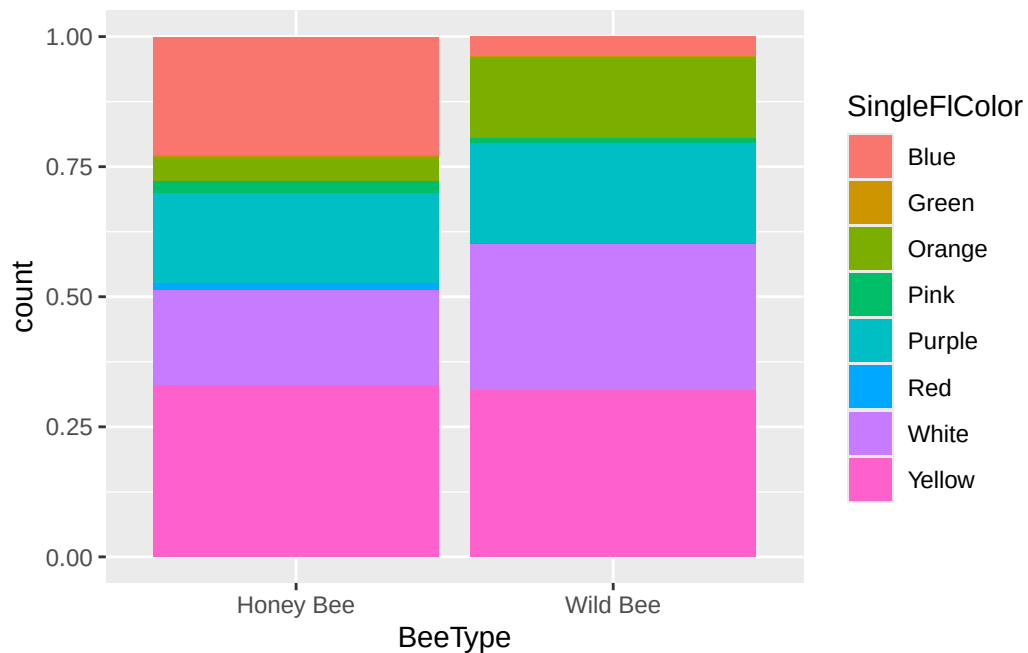
```
[1] "Apidae"      "Halictidae"  "Megachilidae" "Colletidae"
```

Exploratory Graphs

```
# Exploratory Graph based on Order and Flower Color
ggplot(Fl_Color, aes(PollinatorOrder, fill = SingleFlColor)) +
  # make a stacked bar plot to see differences in percent visitation frequency among orders
  geom_bar(position = "fill")
```



```
# Exploratory Graph based on Bee Type and Flower Shape
ggplot(Fl_ColorSp, aes(BeeType, fill = SingleFlColor)) +
  # make a stacked bar plot to see differences in percent visitation frequency among orders
  geom_bar(position = "fill")
```



Contingency Tables

```
# Create contingency table for color and order for easier analysis
ColorOrder <- table(
  Fl_Color$SingleFlColor, Fl_Color$PollinatorOrder)
# print to make sure it works
print(ColorOrder)
```

	Diptera	Hymenoptera	Lepidoptera
Blue	2	107	1
Green	3	2	0
Orange	8	105	3
Pink	1	13	0
Purple	6	172	12
Red	4	7	0
White	137	225	18
Yellow	41	303	6

```
# easy frequentist statistical analysis to see potential alongside graph
c <- chisq.test(ColorOrder)
```

Warning in chisq.test(ColorOrder): Chi-squared approximation may be incorrect

```
print(c)
```

Pearson's Chi-squared test

```
data: ColorOrder
X-squared = 182.92, df = 14, p-value < 2.2e-16
```

```
# Create contingency table for color and order for easier analysis
ColorBee <- table(
  Fl_ColorSp$SingleFlColor, Fl_ColorSp$BeeType)
# print to make sure it works
print(ColorBee)
```

	Honey Bee	Wild Bee
Blue	86	21
Green	1	1
Orange	18	87
Pink	8	5
Purple	65	107
Red	5	2
White	69	156
Yellow	124	179

```
# easy frequentist statistical analysis to see potential alongside graph
d <- chisq.test(ColorBee)
```

Warning in chisq.test(ColorBee): Chi-squared approximation may be incorrect

```
print(d)
```

Pearson's Chi-squared test

```
data: ColorBee
X-squared = 109.37, df = 7, p-value < 2.2e-16
```

Testing brms models for bayesian stats Tube Flower

```
# Create new dataframe for just tubular flowers
Tube_Fl_Order <- Fl_Shape_Final %>%
  filter(FlowerShape == "Tube")
# Print dataframe to see if it workde correctly
Tube_Fl_Order
```

```
# A tibble: 3 x 5
  PollinatorOrder FlowerShape      n total  prop
    <chr>          <chr>    <int> <int> <dbl>
1 Diptera        Tube        36   232 0.155
2 Hymenoptera    Tube       421  1118 0.377
3 Lepidoptera    Tube        22    50 0.44
```

```
m.Tube_Fl_Order <- brm(data = Tube_Fl_Order,
  # Made a beta_binomial distribution and used link = "identity" to not have to back transf
  family = beta_binomial(link = "identity", link_phi = "identity"),
  # categorical model using pollinator order
  n | trials(total) ~ 0 + PollinatorOrder,
  # iterations and warmup alongside number of chains
  iter = 4000, warmup = 2000, chains = 4, cores = 4,
  # setting priors reletively broadly
  prior = c(prior(beta(2, 6), class = "b", dpar = "mu", lb = 0, ub = 1),
    prior(exponential(0.001), class = "phi", lb = 0)),
  file = "output/m.Tube_Fl_Order"
)

# viewing results of model
summary(m.Tube_Fl_Order)
```

```
Family: beta_binomial
Links: mu = identity
Formula: n | trials(total) ~ 0 + PollinatorOrder
Data: Tube_Fl_Order (Number of observations: 3)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000
```

Regression Coefficients:

Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS
----------	-----------	----------	----------	------	----------

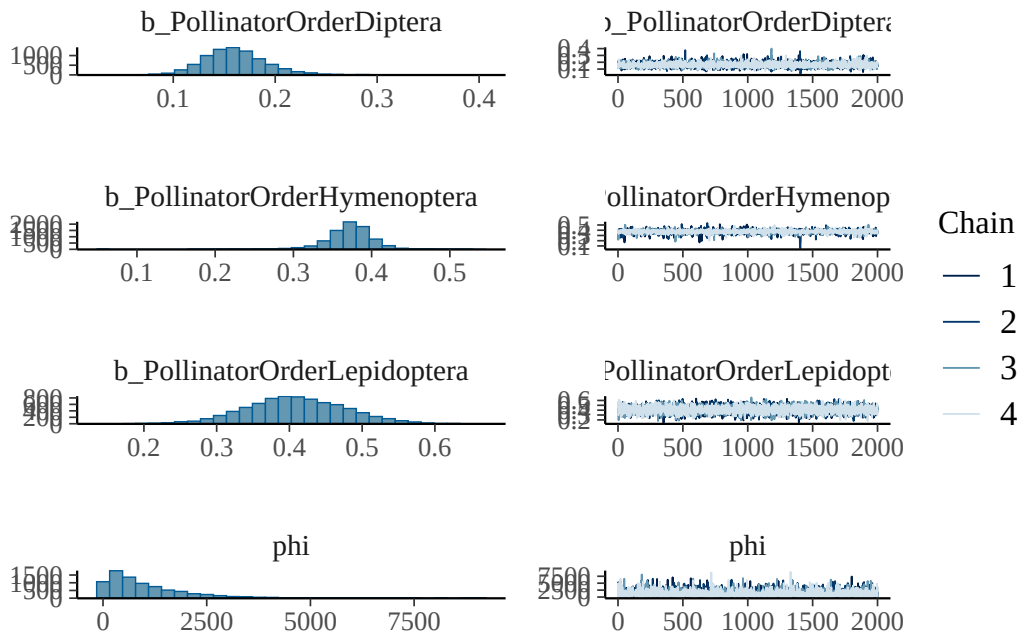
PollinatorOrderDiptera	0.16	0.03	0.11	0.23	1.00	7038
PollinatorOrderHymenoptera	0.37	0.03	0.30	0.43	1.00	4858
PollinatorOrderLepidoptera	0.41	0.07	0.28	0.54	1.00	7339
Tail_ESS						
PollinatorOrderDiptera	4868					
PollinatorOrderHymenoptera	2840					
PollinatorOrderLepidoptera	5481					

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	1043.08	1017.61	38.73	3759.52	1.00	3778	3402

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
# show posterior plots and chains to assess model
plot(m.Tube_Fl_Order)
```



```
m.Tube_Fl_Order1 <-
  brm(data = Tube_Fl_Order, # Give the model the data
      # Chose a binomial distribution
```

```

    family = binomial(link = "logit"),
    # Specified the model
    n | trials(total) ~ 0 + PollinatorOrder,
    # parameters set with default priors
    iter = 4000, warmup = 1000, chains = 4, cores = 4,
    file = "output/m.Tube_Fl_Order1"
  )

# viewing results of model
summary(m.Tube_Fl_Order1)

```

```

Family: binomial
Links: mu = logit
Formula: n | trials(total) ~ 0 + PollinatorOrder
Data: Tube_Fl_Order (Number of observations: 3)
Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
       total post-warmup draws = 12000

```

Regression Coefficients:

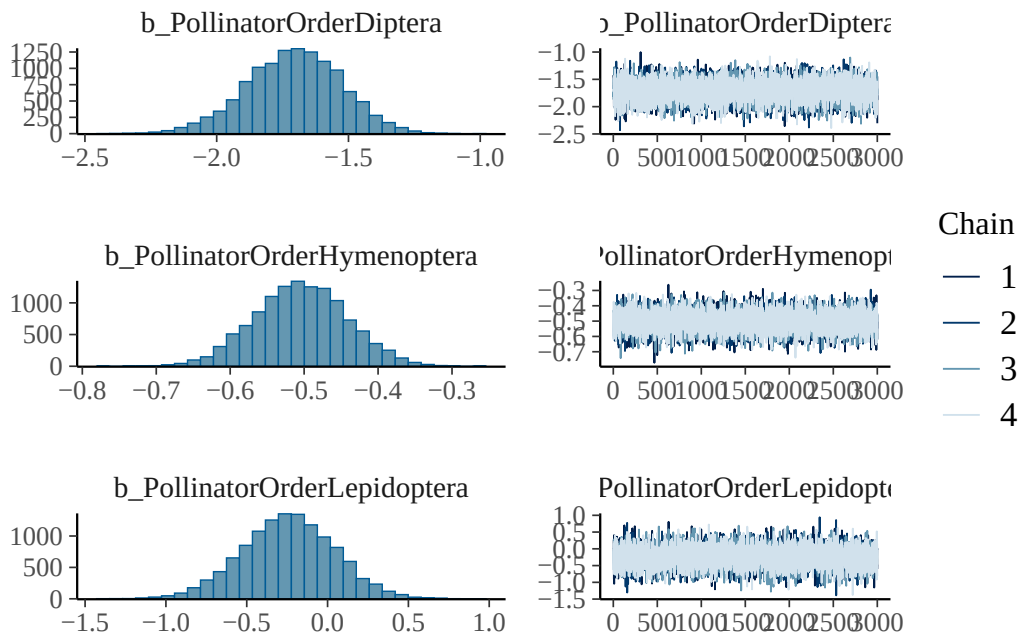
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
PollinatorOrderDiptera	-1.71	0.18	-2.07	-1.36	1.00	13052
PollinatorOrderHymenoptera	-0.50	0.06	-0.63	-0.38	1.00	11714
PollinatorOrderLepidoptera	-0.24	0.29	-0.81	0.31	1.00	12296
	Tail_ESS					
PollinatorOrderDiptera	8318					
PollinatorOrderHymenoptera	8804					
PollinatorOrderLepidoptera	8654					

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```

# show posterior plots and chains to assess model
plot(m.Tube_Fl_Order1)

```



```
# assess model accuracies to see which is better
waic(m.Tube_Fl_Order)
```

Warning:

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

Computed from 8000 by 3 log-likelihood matrix.

	Estimate	SE
elpd_waic	-12.0	1.8
p_waic	1.7	0.1
waic	24.0	3.6

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

```
waic(m.Tube_Fl_Order1)
```

Warning:

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

Computed from 12000 by 3 log-likelihood matrix.

	Estimate	SE
elpd_waic	-11.1	1.4
p_waic	1.5	0.0
waic	22.1	2.7

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

```
# assess model accuracies
loo(m.Tube_Fl_Order, moment_match = TRUE)
```

Computed from 8000 by 3 log-likelihood matrix.

	Estimate	SE
elpd_loo	-14.0	1.8
p_loo	3.7	0.1
looic	28.1	3.7

MCSE of elpd_loo is 0.1.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.3, 0.5]).

All Pareto k estimates are good (k < 0.7).

See help('pareto-k-diagnostic') for details.

```
loo(m.Tube_Fl_Order1, moment_match = TRUE)
```

Error in validate_ll(log_ratios) :

All input values must be finite or -Inf.

Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for d

Warning: Found 3 observations with a pareto_k > 0.7 in model
'm.Tube_Fl_Order1'. We recommend to set 'reloo = TRUE' in order to calculate
the ELPD without the assumption that these observations are negligible. This
will refit the model 3 times to compute the ELPDs for the problematic
observations directly.

Computed from 12000 by 3 log-likelihood matrix.

	Estimate	SE
elpd_loo	-16.3	2.4
p_loo	6.8	2.6
looic	32.7	4.9

MCSE of elpd_loo is NA.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 0.5]).

Pareto k diagnostic values:

		Count	Pct.	Min. ESS
(-Inf, 0.7]	(good)	0	0.0%	<NA>
(0.7, 1]	(bad)	3	100.0%	<NA>
(1, Inf)	(very bad)	0	0.0%	<NA>

See help('pareto-k-diagnostic') for details.

Testing brms models for bayesian stats Dish Flower

```
# Create new dataframe for just dish flowers
Dish_Fl_Order <- Fl_Shape_Final %>%
  filter(FlowerShape == "Dish")
# Print dataframe to see if it workde correctly
Dish_Fl_Order
```

```
# A tibble: 3 x 5
  PollinatorOrder FlowerShape      n total  prop
  <chr>           <chr>    <int> <int> <dbl>
1 Diptera        Dish      163   232 0.703
2 Hymenoptera    Dish      563  1118 0.504
3 Lepidoptera    Dish       22    50 0.44
```

```
m.Dish_Fl_Order <- brm(data = Dish_Fl_Order,
  # Made a beta_binomial distribution and used link = "identity" to not have to back transf
  family = beta_binomial(link = "identity", link_phi = "identity"),
  # categorical model using pollinator order
  n | trials(total) ~ 0 + PollinatorOrder,
  # iterations and warmup alongside number of chains
  iter = 4000, warmup = 2000, chains = 4, cores = 4,
```

```

# setting priors reletively broadly
prior = c(prior(beta(2, 6), class = "b", dpar = "mu", lb = 0, ub = 1),
          prior(exponential(0.001), class = "phi", lb = 0)),
file = "output/m.Dish_Fl_Order"
)

# viewing results of model
summary(m.Dish_Fl_Order)

```

```

Family: beta_binomial
Links: mu = identity
Formula: n | trials(total) ~ 0 + PollinatorOrder
Data: Dish_Fl_Order (Number of observations: 3)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
PollinatorOrderDiptera	0.66	0.07	0.45	0.75	1.00	2253
PollinatorOrderHymenoptera	0.49	0.05	0.33	0.56	1.00	2541
PollinatorOrderLepidoptera	0.40	0.07	0.25	0.55	1.00	3542
	Tail_ESS					
PollinatorOrderDiptera	1025					
PollinatorOrderHymenoptera	1054					
PollinatorOrderLepidoptera	2500					

Further Distributional Parameters:

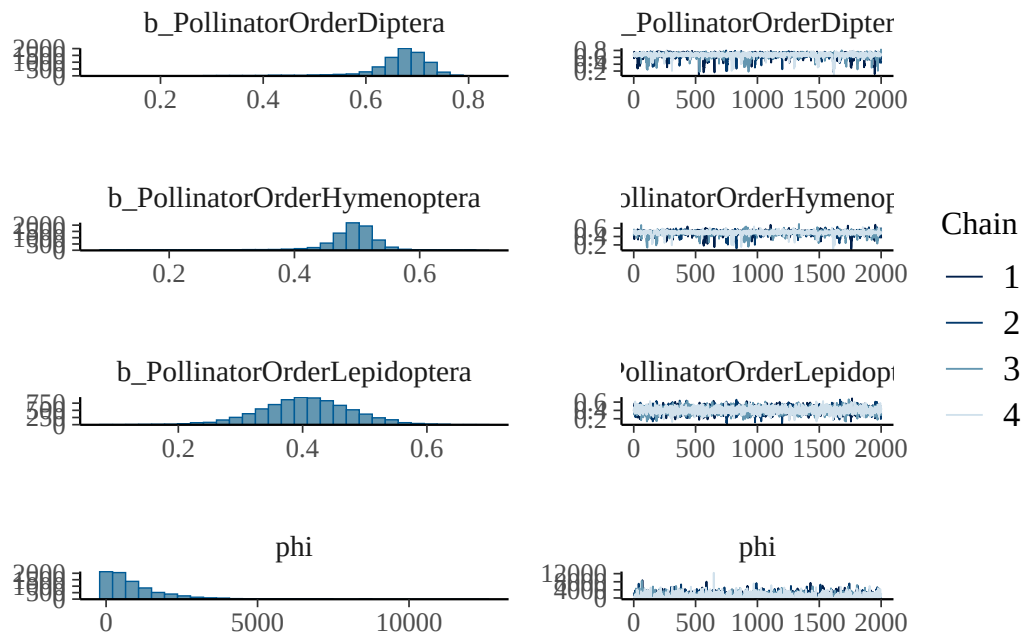
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	911.97	994.44	8.37	3612.32	1.00	1430	1049

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```

# show posterior plots and chains to assess model
plot(m.Dish_Fl_Order)

```



```
m.Dish_Fl_Order1 <-
  brm(data = Dish_Fl_Order, # Give the model the data
    # Chose a binomial distribution
    family = binomial(link = "logit"),
    # Specified the model
    n | trials(total) ~ 0 + PollinatorOrder,
    # parameters set with default priors
    iter = 4000, warmup = 1000, chains = 4, cores = 4,
    file = "output/m.Dish_Fl_Order1"
  )

# viewing results of model
summary(m.Dish_Fl_Order1)
```

```
Family: binomial
Links: mu = logit
Formula: n | trials(total) ~ 0 + PollinatorOrder
Data: Dish_Fl_Order (Number of observations: 3)
Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
       total post-warmup draws = 12000

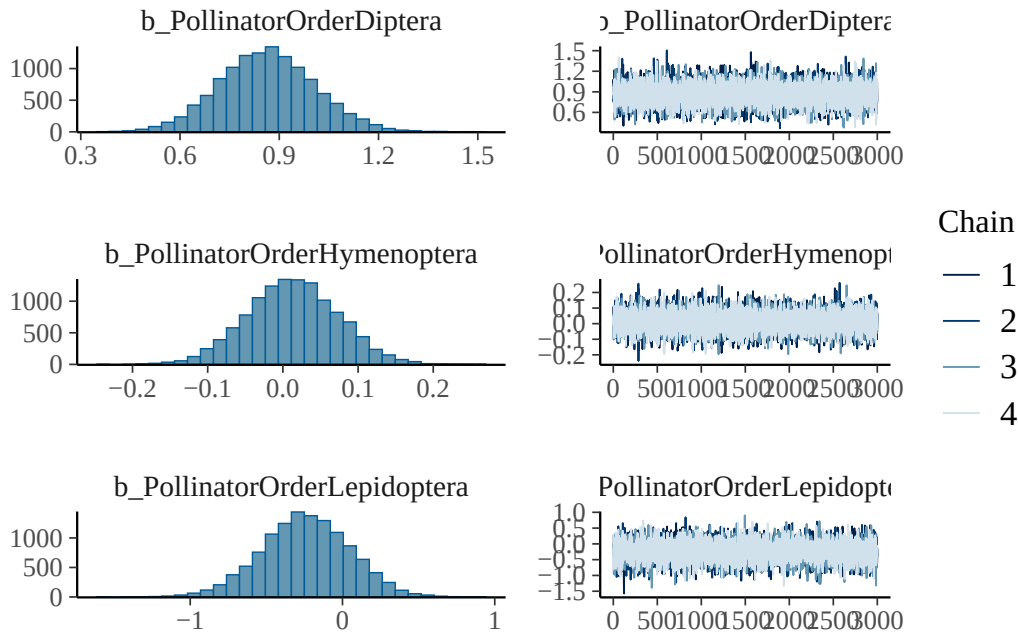
Regression Coefficients:
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
PollinatorOrderDiptera	0.86	0.15	0.58	1.15	1.00	12380
PollinatorOrderHymenoptera	0.01	0.06	-0.10	0.13	1.00	13674
PollinatorOrderLepidoptera	-0.24	0.29	-0.82	0.31	1.00	12464

	Tail_ESS
PollinatorOrderDiptera	9182
PollinatorOrderHymenoptera	9120
PollinatorOrderLepidoptera	8707

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
# show posterior plots and chains to assess model
plot(m.Dish_Fl_Order1)
```



```
# I am having a terrible time with these categorical models. get me out!!!! They don't make
```

```
# assess model accuracies to see which is better
waic(m.Dish_Fl_Order)
```

Warning:

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

Computed from 8000 by 3 log-likelihood matrix.

	Estimate	SE
elpd_waic	-13.4	2.1
p_waic	2.5	0.4
waic	26.8	4.2

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

```
waic(m.Dish_Fl_Order1)
```

Warning:

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

Computed from 12000 by 3 log-likelihood matrix.

	Estimate	SE
elpd_waic	-11.4	1.4
p_waic	1.6	0.0
waic	22.9	2.7

3 (100.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.

```
# assess model accuracies  
loo(m.Dish_Fl_Order, moment_match = TRUE)
```

Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.

Warning: Found 1 observations with a pareto_k > 0.7 in model 'm.Dish_Fl_Order'. We recommend to set 'reloo = TRUE' in order to calculate the ELPD without the assumption that these observations are negligible. This will refit the model 1 times to compute the ELPDs for the problematic observations directly.

Computed from 8000 by 3 log-likelihood matrix.

	Estimate	SE
elpd_loo	-17.2	3.3

```
p_loo          6.4 2.9
looic          34.5 6.6
```

MCSE of elpd_loo is NA.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.2, 0.3]).

Pareto k diagnostic values:

		Count	Pct.	Min. ESS
(-Inf, 0.7]	(good)	2	66.7%	154
(0.7, 1]	(bad)	1	33.3%	<NA>
(1, Inf)	(very bad)	0	0.0%	<NA>

See help('pareto-k-diagnostic') for details.

```
loo(m.Dish_Fl_Order1, moment_match = TRUE)
```

Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.

Warning: Found 3 observations with a pareto_k > 0.7 in model 'm.Dish_Fl_Order1'. We recommend to set 'reloo = TRUE' in order to calculate the ELPD without the assumption that these observations are negligible. This will refit the model 3 times to compute the ELPDs for the problematic observations directly.

Computed from 12000 by 3 log-likelihood matrix.

```
      Estimate SE
elpd_loo   -17.1 4.6
p_loo       7.3 3.3
looic      34.2 9.2
```

MCSE of elpd_loo is NA.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 0.6]).

Pareto k diagnostic values:

		Count	Pct.	Min. ESS
(-Inf, 0.7]	(good)	0	0.0%	<NA>
(0.7, 1]	(bad)	3	100.0%	<NA>
(1, Inf)	(very bad)	0	0.0%	<NA>

See help('pareto-k-diagnostic') for details.